EUCLID · BOOK I · PROPOSITION 20

# Turn two sides into one longer straight length

Extend a side and copy the other onto it. The earlier angle–side comparison then proves the triangle inequality without assuming that a straight route is shortest.

## THE PROPOSITION

**GREEK** Ἑλληνικά

Παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

**LATIN** Latine

In quouis triangulo duo latera reliquo maiora sunt quoquo modo coniuncta.

**ENGLISH** reading translation

In any triangle, any two sides taken together are greater than the remaining side.

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1 Name all three comparisons

**GREEK** Ἑλληνικά

Ἔστω γὰρ τρίγωνον τὸ ΑΒΓ· λέγω, ὅτι τοῦ ΑΒΓ τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν ΒΑ, ΑΓ τῆς ΒΓ, αἱ δὲ ΑΒ, ΒΓ τῆς ΑΓ, αἱ δὲ ΒΓ, ΓΑ τῆς ΑΒ.

**LATIN** Latine

Sit enim triangulus ΑΒΓ. dico, in triangulo ΑΒΓ duo latera reliquo maiora esse quoquo modo coniuncta, ΒΑ + ΑΓ > ΒΓ, ΑΒ + ΒΓ > ΑΓ, ΒΓ + ΓΑ > ΑΒ.

**ENGLISH** reading translation

Let ΑΒΓ be a triangle. I say that any two sides together exceed the remaining side: ΒΑ and ΑΓ exceed ΒΓ; ΑΒ and ΒΓ exceed ΑΓ; ΒΓ and ΓΑ exceed ΑΒ.

## 2 Continue BA beyond A

**GREEK** Ἑλληνικά

Διήχθω γὰρ ἡ ΒΑ ἐπὶ τὸ Δ σημεῖον,

**LATIN** Latine

educatur enim BA ad Δ punctum,

**ENGLISH** reading translation

For let BA be continued to the point Δ,

## 3 Copy ΓA to AΔ, then join ΔΓ

**GREEK** Ἑλληνικά

καὶ κείσθω τῇ ΓΑ ἴση ἡ ΑΔ, καὶ ἐπεζεύχθω ἡ ΔΓ.

**LATIN** Latine

et ponatur AΔ = ΓA, et ducatur ΔΓ.

**ENGLISH** reading translation

and let AΔ be laid off equal to ΓA, and let ΔΓ be joined.

## 4 Use the small isosceles triangle

**GREEK** Ἑλληνικά

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔΑ τῇ ΑΓ, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ΑΔΓ τῇ ὑπὸ ΑΓΔ·

**LATIN** Latine

iam quoniam ΔA = AΓ, erit etiam ∠ AΔΓ = AΓΔ.

**ENGLISH** reading translation

Since ΔA equals AΓ, angle AΔΓ also equals AΓΔ.

## 5 Compare the whole angle with its part

**GREEK** Ἑλληνικά

μείζων ἄρα ἡ ὑπὸ ΒΓΔ τῆς ὑπὸ ΑΔΓ.

**LATIN** Latine

itaque ∠ ΒΓΔ > ΑΔΓ.

**ENGLISH** reading translation

Therefore ΒΓΔ is greater than ΑΔΓ.

## 6 Use I.19 in the larger triangle

**GREEK** Ἑλληνικά

καὶ ἐπεὶ τρίγωνόν ἐστι τὸ ΔΓΒ μείζονα ἔχον τὴν ὑπὸ ΒΓΔ γωνίαν τῆς ὑπὸ ΒΔΓ, ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, ἡ ΔΒ ἄρα τῆς ΒΓ ἐστι μείζων.

**LATIN** Latine

et quoniam triangulus est ΔΓΒ maiorem habens angulum ΒΓΔ angulo ΒΔΓ, sub maiore autem angulo maius latus subtendit, erit ΔΒ > ΒΓ.

**ENGLISH** reading translation

Since triangle ΔΓΒ has angle ΒΓΔ greater than ΒΔΓ, and the side opposite the greater angle is greater, ΔΒ is greater than ΒΓ.

## 7 Replace ΔΑ with its equal ΑΓ

**GREEK** Ἑλληνικά

ἴση δὲ ἡ ΔΑ τῇ ΑΓ· μείζονες ἄρα αἱ ΒΑ, ΑΓ τῆς ΒΓ·

**LATIN** Latine

uerum ΔΑ = ΑΓ. itaque ΒΑ + ΑΓ > ΒΓ.

**ENGLISH** reading translation

But ΔΑ equals ΑΓ. Therefore ΒΑ and ΑΓ together are greater than ΒΓ.

## 8 Construct the other two comparisons

**GREEK** Ἑλληνικά

ὁμοίως δὴ δείξομεν, ὅτι καὶ αἱ μὲν ΑΒ, ΒΓ τῆς ΓΑ μείζονές εἰσιν, αἱ δὲ ΒΓ, ΓΑ τῆς ΑΒ.

**LATIN** Latine

similiter demonstrabimus, esse etiam AB + BΓ > ΓA et BΓ + ΓA > AB.

**ENGLISH** reading translation

Similarly we shall show that AB and BΓ together exceed ΓA, and that BΓ and ΓA together exceed AB.

## 9 Conclude the triangle inequality

**GREEK** Ἑλληνικά

Παντὸς ἄρα τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

**LATIN** Latine

Ergo in quouis triangulo duo latera reliquo maiora sunt quoquo modo coniuncta; quod erat demonstrandum.

**ENGLISH** reading translation

Therefore, in any triangle, any two sides taken together exceed the remaining side. This is what had to be shown.

Stoicheia · Euclid, Elements I.1–20. Greek and Latin: J. L. Heiberg, Euclidis opera omnia, vol. I (Leipzig, 1883), pp. 2–51; public domain. Latin is Heiberg’s nineteenth-century translation. English reading and notes: Stoicheia, 2026, MIT. Source and edition notes: antonsoo.github.io/stoicheia/